13/10/26

LOW VOLTAGE : BSPD PROJECT

BSPD : Brake System Plausibility Device

↳ Shutdown system of the car
↳ Non-programmable circuit → opens the shutdown circuit

The main idea is manual shutdown of the car when the following conditions are met:

→ Demand for high break for 300ms
→ Tractive system current is at a level where 5kW of electrical power in the DC circuit is being delivered to the motor(s) at the nominal battery charge

* It must also "open" the shutdown circuit when there is an open / short circuit in any input

→ Must not send power to the Motor(s) of the vehicle

→ For the test we must provide a wire / coil to simulate Tractive System

→ Provide connection points that are two 4mm shrouded bananajacks (CS+ and CS-) (not black / red)

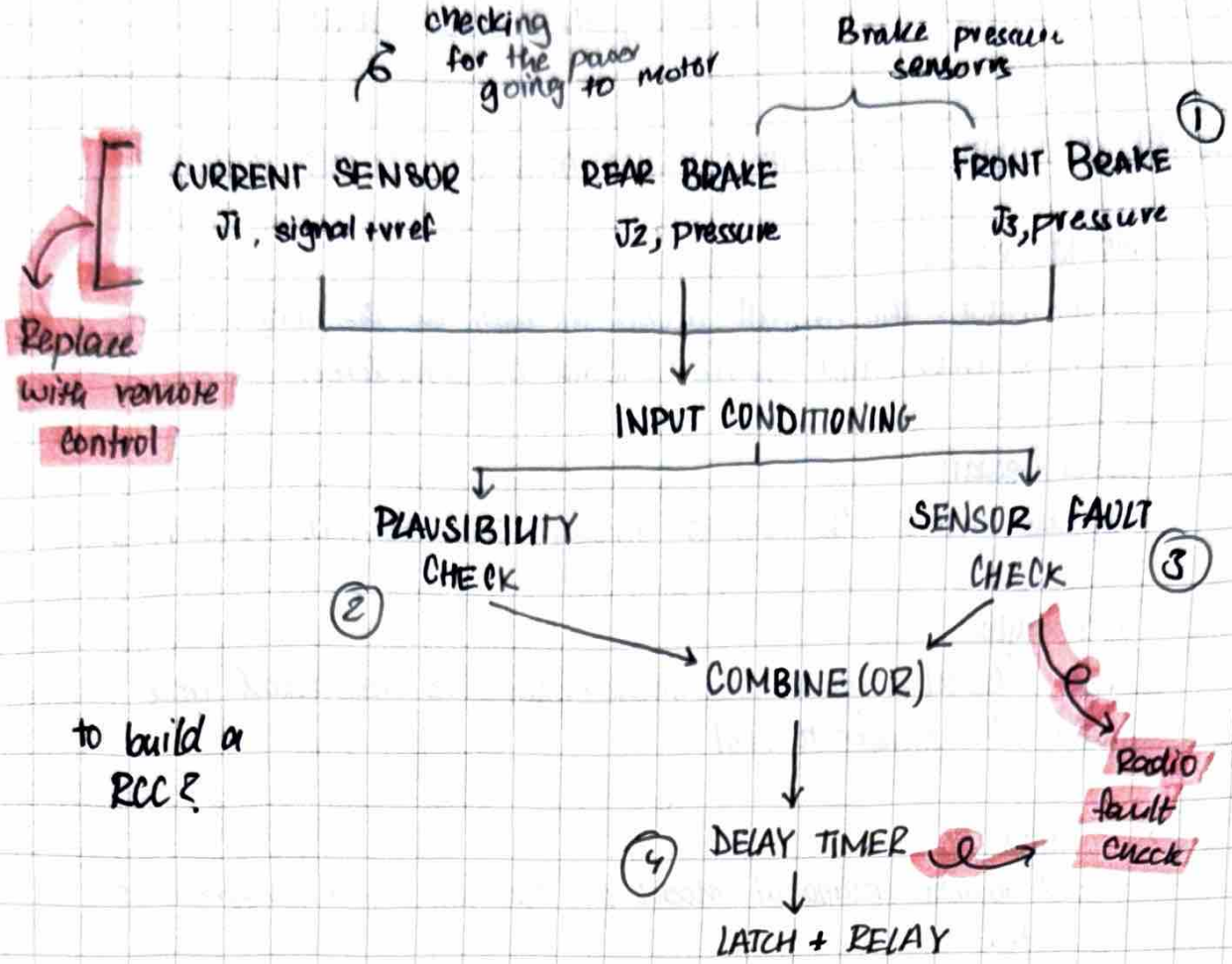
→ Locations possible : On the vehicle, On an adaptor harness or breakout box

→ Components

→ Brake Pressure
→ Current Sensor : → window comparator (LM339)

→ Delay

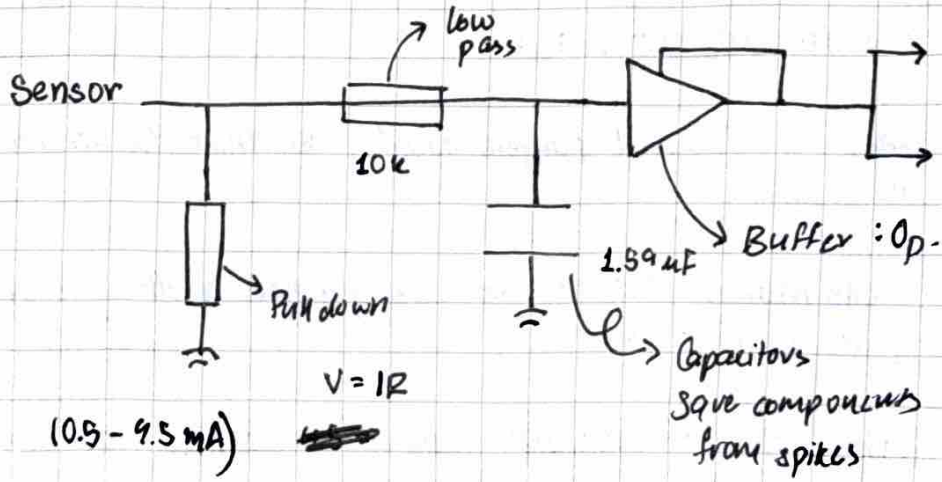
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QUESTIONS

1. What happens when we open the shut-down circuit?
2. what kind of output do the sensors give is it CAN buses?
   - Can't be this circuit is analog.

(1) Sensor detection



CIRCUITS I FOUND INTERESTING

~~RES System (Remote Emergency Stop Hardware)~~ → is german

- → 430 MHz ~~to 440 MHz~~ : remote controlled (not necessary for testing purposes)
- → ~~Open~~ the shut down circuit

→ We need RSS, not RES.
↳ Remote Stop System

1) Pulldown (R1 → R4)

R_1, R_2 → 220k >> current sensor

R_3, R_4 → 100k >> Brake sensors

Current Sensors >> which model ave we working on

$P = V \times I$

6kW =

~~R~~ BREK SENSORS >> 0.5 → 4.5V
grows with pressure on the brake

$0.5 = I \times 100k$

$I = \frac{0.5}{100000}$

$I = 0.5 \times 10^{-4}$

$= 0.5\ mA$

$4.5 = I \times 100k$

$I = 4.5 \times 10^{-4}$

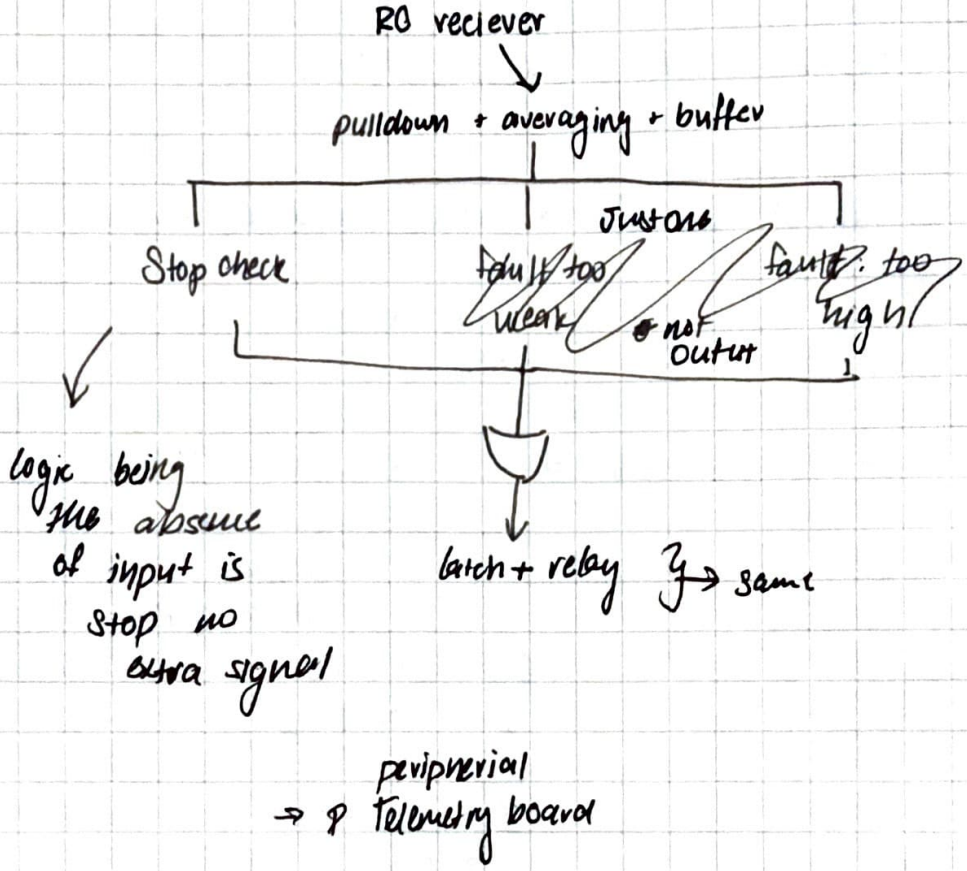
$= 4.5\ mA$

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RSS specifications

→ NO Delay
→ Unobstructed antennas
→
→ Stop button: open shutdown circuit
→ Go button: preselected driverless motion may start

→ Remote System has
  > Vehicle module
  > Control box
  > Remote Stop Relay



peripherial
→ P telemetry board

CHOOSING A RADIO

What specifications would we need?

- → update rate: 10 – 30 ms
- → range:
- → implementations style: easier side
- → cost: 200 or less (the lesser the better)

Options:

	LoRA	FSK	Arduino + radio module	ESP32
Imple mentation	+ ESP32?	CRC	Can ~~be~~ taping and all doesn't work that well	seems easier to implement?

NEXT STEPS

1. Pick an esp32 with external antenna and CAN peripheral
   - ↳ needs to have U, UE, or IE in the name
   - ↗ almost all ESP32 have that
   - CAN = Controlled Area Network peripheral → manages serial data comms over CAN buses

ESP32 NOW
- is connectionless wireless protocol that runs on the chip to send short data packets.
- is supported by:
  - ESP8266
  - ESP32
  - ESP32-S
  - ESP32-C

⇒ Classic CAN vs CAN FD
- Classic CAN → 8 bytes per message
- CAN FD → flexible data

→ Need a transciever → FS schematics
use TI ISO1044BD

PICK: ESP32-S3-WROOM-1U » native USB
→ newest
→ lots of examples online

———— • * * * • ————

IMPLEMENTING ANY NEEDED EXTERNAL COMPONENTS FOR IT

> Transciever : the team's

* > CAN Bus

> Converter (12V → 5V) >> VCU circuit has this
↓
copy the stuff from there

> 3.3V LDO

SECTION 3 | CAMPUS 17

- StaticBody 2D

Essentially does not
bodies

used for : Platformer

- Area 2D

detects collision, keeps
can alter gravity,

- Collison Polygon 2D

polygon shape on

Warning: non uniform

- Ray Cast 2D

kind of like a po
finds the closest

- Camera 2D

forces the screen

- Singletons

Anything that

Skeletons, decid